

In this video, we build a GitHub Actions CI/CD pipeline for a multi-build variant Android app.

`.github/workflows/android-release-build.yml`

<https://x.com/anilvdes...>

Official WebSite: <https://codetutorhub.dev/>

GitHub link to code: <https://github.com/Ani>

Original file line number	Diff line number	Diff line change
---------------------------	------------------	------------------

@@ -0,0 +1,129 @@		
-------------------	--	--

# — Android CI — Scalable Secret Strategy		—————
---	--	-------

This workflow lives on `main` (so GitHub discovers it in the Actions tab)

but checks out whichever branch you specify in the inputs.

#

Secret injection follows the same classification as build.gradle.kts:

COMMON → ANALYTICS_SDK_KEY (always injected)

ENVIRONMENT-SPECIFIC → BACKEND_TOKEN_<ENV> (all three, always)

TIER-SPECIFIC → AD_SDK_KEY_<TIER> (both, always)

RELEASE-ONLY → RELEASE_SIGNING_* (only when buildType=release)

#

 IMPORTANT: Gradle evaluates ALL productFlavors at configuration time,

so all common/env/tier secrets must be present for every build.

Only signing secrets are truly conditional (use resolveSecretOrNull).

#

No giant if-else. The variant name encodes the env, tier, and buildType.

name: Android Manual Variant Build

on:

workflow_dispatch:

inputs:

branch:

description: 'Branch to build from'

required: true

type: string

default: 'live-coding'

variant:

description: 'Variant to build'

required: true

type: choice

options:

- qaFreeDebug

- qaPaidDebug

- stagingFreeDebug
- stagingPaidDebug
- prodFreeRelease
- prodPaidRelease

jobs:

build:

runs-on: ubuntu-latest

steps:

- name: Checkout code

uses: actions/checkout@v5

with:

ref: \${{ inputs.branch }}

- name: Derive env, tier, and buildType from variant name

id: classify

run: |

VARIANT="\${{ inputs.variant }}"

echo "variant=\$VARIANT"

Extract environment (first segment: qa, staging, or prod)

if [["\$VARIANT" == qa*]]; then

ENV="QA"

elif [["\$VARIANT" == staging*]]; then

ENV="STAGING"

elif [["\$VARIANT" == prod*]]; then

ENV="PROD"

fi

Extract tier (middle segment: Free or Paid)

if [["\$VARIANT" == *Free*]]; then

TIER="FREE"

elif [["\$VARIANT" == *Paid*]]; then

TIER="PAID"

fi

Extract build type (last segment: Debug or Release)

if [["\$VARIANT" == *Debug]]; then

BUILD_TYPE="debug"

elif [["\$VARIANT" == *Release]]; then

BUILD_TYPE="release"

fi

{

```

    echo "env=$ENV"
    echo "tier=$TIER"
    echo "buildType=$BUILD_TYPE"
} >> "$GITHUB_OUTPUT"

echo "— Classification —"
echo " Environment : $ENV"
echo " Tier       : $TIER"
echo " Build Type  : $BUILD_TYPE"
- name: Set up JDK 21
  uses: actions/setup-java@v5
  with:
    distribution: temurin
    java-version: '21'

- name: Build variant
  env:
    # — COMMON (always) —————
    ANALYTICS_SDK_KEY: ${ secrets.ANALYTICS_SDK_KEY }

    # — ENVIRONMENT-SPECIFIC (all three — Gradle configures —
    # every flavor at configuration time, regardless of which
    # variant you're building)
    BACKEND_TOKEN_QA: ${ secrets.BACKEND_TOKEN_QA }
    BACKEND_TOKEN_STAGING: ${ secrets.BACKEND_TOKEN_STAGING }
    BACKEND_TOKEN_PROD: ${ secrets.BACKEND_TOKEN_PROD }

    # — TIER-SPECIFIC (both — same reason as above) —————
    AD_SDK_KEY_FREE: ${ secrets.AD_SDK_KEY_FREE }
    AD_SDK_KEY_PAID: ${ secrets.AD_SDK_KEY_PAID }

    # — RELEASE-ONLY (signing — only when buildType=release) —
    # These are the one category that IS truly conditional:
    # signingConfigs uses resolveSecretOrNull, so empty is OK.
    RELEASE_SIGNING_STORE_FILE:  ${ steps.classify.outputs.buildType == 'release'
    && secrets.RELEASE_SIGNING_STORE_FILE  || " }
    RELEASE_SIGNING_STORE_PASSWORD: ${ steps.classify.outputs.buildType ==
    'release' && secrets.RELEASE_SIGNING_STORE_PASSWORD || " }
    RELEASE_SIGNING_KEY_ALIAS:    ${ steps.classify.outputs.buildType == 'release' &&
    secrets.RELEASE_SIGNING_KEY_ALIAS    || " }
    RELEASE_SIGNING_KEY_PASSWORD: ${ steps.classify.outputs.buildType ==
    'release' && secrets.RELEASE_SIGNING_KEY_PASSWORD || " }
  run: |
    chmod +x ./gradlew

```

```

    VARIANT="${{ inputs.variant }}"
    TASK_NAME="assemble${VARIANT^}"
    echo "Building: ./gradlew $TASK_NAME"
    echo " Secrets injected for: env=${{ steps.classify.outputs.env }}, tier=${{
steps.classify.outputs.tier }}, buildType=${{ steps.classify.outputs.buildType }}"
    ./gradlew "$TASK_NAME"
  - name: Upload APK
    uses: actions/upload-artifact@v5
    with:
      name: ${{ inputs.variant }}-apk
      path: app/build/outputs/apk/**/*.*.apk

```

github/workflows/android-main-build.yml

github/workflows/android-pr-checks.yml and

.github/workflows/android-release-build.yml

-0,0 +1,56 @@

name: Android Release Build

on:

workflow_dispatch:

inputs:

variant:

description: 'Release variant to build'

required: true

type: choice

options:

- prodFreeRelease
- prodPaidRelease

permissions:

contents: read

jobs:

signed-release:

name: Trusted release build

runs-on: ubuntu-latest

environment: production

steps:

- name: Checkout code

uses: actions/checkout@v5

- name: Set up JDK 21

uses: actions/setup-java@v5

with:

distribution: temurin

java-version: '21'

- name: Build signed release variant

env:

ANALYTICS_SDK_KEY: \${ secrets.ANALYTICS_SDK_KEY }

BACKEND_TOKEN_QA: \${ secrets.BACKEND_TOKEN_QA }

BACKEND_TOKEN_STAGING: \${ secrets.BACKEND_TOKEN_STAGING }

BACKEND_TOKEN_PROD: \${ secrets.BACKEND_TOKEN_PROD }

AD_SDK_KEY_FREE: \${ secrets.AD_SDK_KEY_FREE }

AD_SDK_KEY_PAID: \${ secrets.AD_SDK_KEY_PAID }

RELEASE_SIGNING_STORE_FILE: \${ secrets.RELEASE_SIGNING_STORE_FILE }

RELEASE_SIGNING_STORE_PASSWORD: \${ secrets.RELEASE_SIGNING_STORE_PASSWORD }

RELEASE_SIGNING_KEY_ALIAS: \${ secrets.RELEASE_SIGNING_KEY_ALIAS }

RELEASE_SIGNING_KEY_PASSWORD: \${ secrets.RELEASE_SIGNING_KEY_PASSWORD }

run: |

chmod +x ./gradlew

VARIANT="\${ inputs.variant }"

TASK_NAME="assemble\${VARIANT^}"

echo "Building trusted release artifact: ./gradlew \$TASK_NAME"

./gradlew "\$TASK_NAME"

- name: Upload signed release APK

uses: actions/upload-artifact@v5

with:

name: \${ inputs.variant }-signed-apk

path: app/build/outputs/apk/**/*.*.apk